

SE 216 – SOFTWARE PROJECT MANAGEMENT

SOFTWARE MEASUREMENTS DOCUMENT

PROJECT NAME:

Monitoring and Assistance System for At-Home Physiotherapy Exercises

GROUP NUMBER and MEMBERS:

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Questions to identify measurements:

- Is the project progressing according to the planned schedule?
- Does the YOLO Pose algorithm achieve the required 90% accuracy in detecting incorrect movements?
- How quickly can the system detect and warn users about incorrect movements?
- How much effort (staff-hours) are team members spending on development, testing, and documentation?
- Can elderly users navigate the interface and start an exercise session without external help?
- Does the system operate reliably without crashes or data loss during exercise sessions?

Identified measurements:

- Planned task completion date/ Actual task completion date.
- Percentage of incorrectly performed movements correctly detected during physiotherapy sessions (Target: $\geq 90\%$).
- Average response time for detecting and warning incorrect movements in real-time.
- Total staff-hours logged per week for development, testing, and documentation activities.
- A survey-based score (1–5) where users rate how easy it was to understand the on-screen alerts during exercise. /The percentage of elderly users who can successfully start an exercise without any external assistance.
- Percentage of exercise sessions completed without crashes, freezes, or data loss.

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Measurement storage and collection:

What: Task Completion

When: At the end of each task completed

Format: Date

How: Entered by the project manager into the shared project schedule or

What: Correctly identified user errors (YOLO Pose Accuracy)

When: After each algorithm testing session or assisted user trial.

Format: Percentage (%) and integer counts of correct/incorrect detections

How: Recorded by the development team into a performance log or shared testing spreadsheet

What: Weekly staff-hours for development, testing, and documentation

When: At the end of each week (every Sunday)

Format: Integer or real number data

How: Each team member records their individual hours in a shared team schedule

What: User survey scores and task success rates (Usability)

When: After each usability testing session with elderly participants

Format: Integer (1–5 scale) and percentage (%)

How: Collected via digital feedback forms and recorded by the UX/UI tester in the usability log.

What: Incorrect movement response time

When: After each algorithm testing session

Format: Seconds (s)

How: Automatically recorded by the application performance logs.

What: Successful exercise sessions completed without crashes or data loss

When: After each user session

Format: Percentage (%)

How: Automatically recorded through system monitoring logs.

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Measurement Type	Description	Example Measurements
Schedule	Comparison between the planned milestone dates and the actual dates tasks were completed.	Motion Analysis Module: Planned April 15, Completed April 18.
Effort	Total staff-hours logged per week for development, testing, and documentation activities.	28 total staff-hours in Week 4.
AI Performance	The percentage of incorrectly performed movements correctly detected during physiotherapy sessions (Target: $\geq 90\%$).	92% accuracy achieved in algorithm testing.
UI Clarity	A survey-based score (1–5) evaluating how easily elderly users understand on-screen alerts and navigate the system independently.	Average usability score of 4.5 / 5 from elderly users during exercise sessions.
Performance	Average response time of the system in detecting and warning incorrect/correct movements in real-time.	Incorrect movement detected within 1.8 seconds
Reliability	Percentage of exercise sessions completed without crashes, freezes, or data loss.	97% of exercise sessions completed successfully without interruption.